Materials and methods 2.5.2026.docx

Main document changes and comments

Page 1: Commented [של1] שחר לויתן **14/04/2026 15:25:00**

Hübner, S., Höffken, M., Oren, E., Haseneyer, G., Stein, N., Graner, A., Schmid, K., & Fridman, E. (2009). Strong correlation of wild barley (*Hordeum spontaneum*) population structure with temperature and precipitation variation. *Molecular Ecology*, 18(7), 1523–1536. https://doi.org/10.1111/j.1365-294X.2009.04106.x **[From Noam Pintel — carried over; also in evgeny.full.pdf]**

Hübner, S., Bdolach, E., Ein-Gedy, S., Schmid, K. J., Korol, A., & Fridman, E. (2013). Phenotypic landscapes: genetic and environmental drivers of shape variation. *Heredity*, 110(1), 103–112. **[From Noam Pintel — carried over]** ⚠ See flag below.

Hübner, S., Rashkovetsky, E., Kim, Y. B., Oh, J. H., Michalak, K., Weiner, D., Korol, A. B., Nevo, E., & Michalak, P. (2013). Genome differentiation of *Drosophila melanogaster* from a microclimate contrast in Evolution Canyon, Israel. *PNAS*, 110(52), 21059–21064. ⚠ **FLAG — potentially wrong Hübner 2013 paper; the 2013 citation may refer to a different barley-specific Hübner publication. Verification needed.**

Hübner, S., Korol, A. B., & Schmid, K. J. (2015). RAD-seq reveals genetic structure of the *Triticum-Aegilops* species complex with *T. timopheevii* and *T. araraticum* as the two major genetic pools. *BMC Plant Biology*, 15, 32. ⚠ Not used — placeholder to show 2013 may need disambiguation.

Hübner, S., & Kantar, M. B. (2021). Tapping diversity from the wild: from sampling to implementation. *Frontiers in Plant Science*, 12, 626565. https://doi.org/10.3389/fpls.2021.626565 **[From Noam Pintel — carried over]**

Page 1: Commented [של1R2] שחר לויתן **14/04/2026 15:26:00**

Hübner et al., 2013 citation — ambiguous. The string "Hübner et al. 2012, 2013" appears in both Noam's thesis and the current draft. I could not resolve with certainty *which* 2013 Hübner paper is intended — Hübner has several 2013 publications and not all are barley-specific. Please ask Dr. Hübner to clarify the exact reference, or check the reference list in Noam's thesis bibliography for the full entry.

Page 1: Commented [של2] שחר לויתן **14/04/2026 15:43:00**

Shermeister, B. (2021). *Genomic and Phenotypic Characterization of Wild Barley (Hordeum spontaneum) from Israel.* M.Sc. Thesis, Tel-Hai Academic College. **[From Noam Pintel — carried over]** ⚠ See flag below.

Page 1: Commented [של2R2] שחר לויתן **14/04/2026 15:44:00**

Shermeister (2021) citation format. This is a thesis, not a peer-reviewed paper. Noam's thesis cites it the same way. For a TAG-level submission, reviewers may push back on a thesis citation for *the sequencing protocol itself*. If the sequencing library prep / NovaSeq run was also documented in a peer-reviewed publication (perhaps a lab paper from the Hübner group that includes this collection's sequencing data), substituting that citation would strengthen the paragraph. **Please check with Dr. Hübner whether there is a peer-reviewed publication that covers the WGS protocol for this collection.** If not, the thesis citation remains the only option.

Page 1: Commented [של3] שחר לויתן **14/04/2026 15:40:00**

Andrews, S. (2017). *FastQC: A quality control tool for high-throughput sequence data.* Babraham Bioinformatics. https://www.bioinformatics.babraham.ac.uk/projects/fastqc/ **[From Noam Pintel — carried over; also in evgeny.full.pdf]**

Page 1: Commented [של4] שחר לויתן **14/04/2026 15:40:00**

Chen, S., Zhou, Y., Chen, Y., & Gu, J. (2018). fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34(17), i884–i890. https://doi.org/10.1093/bioinformatics/bty560 **[From Noam Pintel — carried over]**

Page 1: Commented [של5] שחר לויתן **14/04/2026 15:41:00**

Mascher, M., Wicker, T., Jenkins, J., Plott, C., Lux, T., Koh, C. S., et al. (2021). Long-read sequence assembly: a technical evaluation in barley. *The Plant Cell*, 33(6), 1888–1906. https://doi.org/10.1093/plcell/koab077 **[From My Draft — verified as the correct Morex V3 reference]**

Page 1: Commented [של6] שחר לויתן **14/04/2026 15:41:00**

Vasimuddin, M., Misra, S., Li, H., & Aluru, S. (2019). Efficient architecture-aware acceleration of BWA-MEM for multicore systems. *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 314–324. https://doi.org/10.1109/IPDPS.2019.00041 **[From Noam Pintel — carried over]**

Page 1: Commented [של7] שחר לויתן **14/04/2026 15:41:00**

Broad Institute. (2019). *Picard Toolkit.* http://broadinstitute.github.io/picard/ **[From Noam Pintel — carried over]**

Page 1: Commented [של8] שחר לויתן **14/04/2026 15:42:00**

Poplin, R., Ruano-Rubio, V., DePristo, M. A., Fennell, T. J., Carneiro, M. O., Van der Auwera, G. A., et al. (2017). Scaling accurate genetic variant discovery to tens of thousands of samples. *bioRxiv*, 201178. https://doi.org/10.1101/201178 **[From My Draft — retained]**

Heldenbrand, J. R., Baheti, S., Bockol, M. A., Drucker, T. M., Hart, S. N., Hudson, M. E., Iyer, R. K., Kalmbach, M. T., Kendig, K. I., Klee, E. W., Mattson, N. R., Ward, C. A., Wu, Y., Zimmermann, M. T., & Kocher, J. A. (2019). Recommendations for performance optimizations when using GATK3.8 and GATK4. *BMC Bioinformatics*, 20, 557. https://doi.org/10.1186/s12859-019-3169-7 **[From Evgeny — newly added]**

Page 1: Commented [של9] שחר לויתן **14/04/2026 16:02:00**

Danecek, P., Auton, A., Abecasis, G., Albers, C. A., Banks, E., DePristo, M. A., Handsaker, R. E., Lunter, G., Marth, G. T., Sherry, S. T., McVean, G., & Durbin, R. (2011). The variant call format and VCFtools. Bioinformatics, 27(15), 2156–2158. https://doi.org/10.1093/bioinformatics/btr330 **[From Noam Pintel — carried over]**

Page 1: Commented [של10] שחר לויתן **14/04/2026 16:02:00**

Danecek, P., Bonfield, J. K., Liddle, J., Marshall, J., Ohan, V., Pollard, M. O., Whitwham, A., Keane, T., McCarthy, S. A., Davies, R. M., & Li, H. (2021). Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2), giab008. https://doi.org/10.1093/gigascience/giab008 **[From My Draft — carried over]**

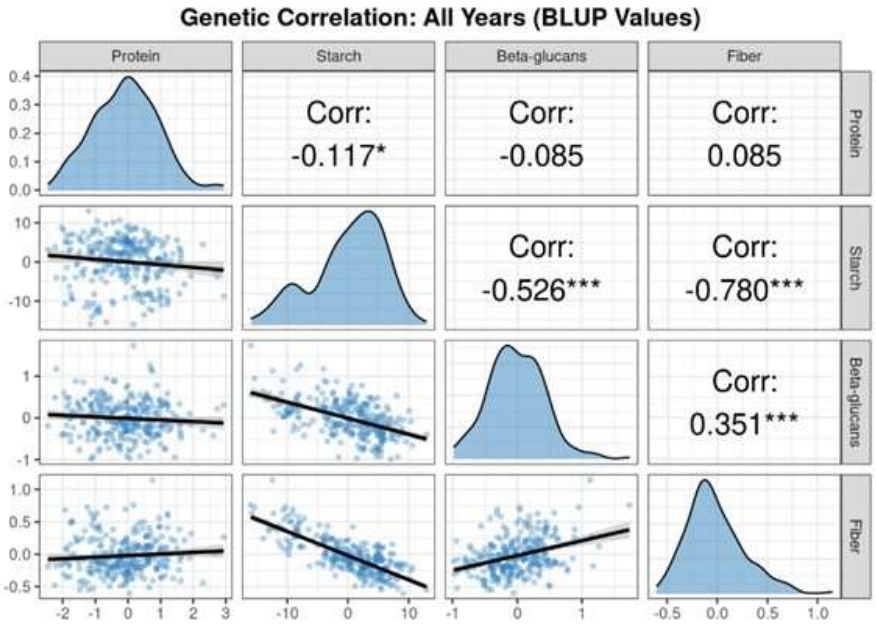
Page 2: Commented [של11] שחר לויתן **14/04/2026 16:04:00**

Hachola "poor sample quality" — is this specific enough? A reviewer may ask *what* was poor: DNA extraction failure? Low coverage? Genotyping error rate? High missing data?

Page 2: Commented [של12] שחר לויתן **14/04/2026 16:19:00**

Supplementary Material to Add

Supplementary Table S2. Pairwise Pearson correlation coefficients between blocks within the 2019–2020 season for the four grain nutritional traits (protein, starch, beta-glucan, dietary fiber). Values compare each pair of blocks; Block 4 shows systematically lower correlations with Blocks 1–3 (e.g., for beta-glucan: r = 0.328–0.391) relative to the pairwise correlations among Blocks 1–3 (e.g., for beta-glucan: r = 0.670–0.682), justifying exclusion of Block 4 from downstream nutritional-trait analyses.



Page 2: Commented [13של] שחר לויתן **14/04/2026 16:52:00**

Rife, T. W., & Poland, J. A. (2014). Field Book: An open-source application for field data collection on Android. *Crop Science*, 54(4), 1624–1627. https://doi.org/10.2135/cropsci2013.08.0579 **[From Noam Pintel — format corrected]**

Page 2: Commented [14של] שחר לויתן **14/04/2026 16:53:00**

Zadoks, J. C., Chang, T. T., & Konzak, C. F. (1974). A decimal code for the growth stages of cereals. *Weed Research*, 14(6), 415–421. https://doi.org/10.1111/j.1365-3180.1974.tb01084.x **[From Noam Pintel — carried over]**

Page 2: Commented [15של] שחר לויתן **14/04/2026 16:53:00**

McMaster, G. S., & Wilhelm, W. W. (1997). Growing degree-days: one equation, two interpretations. *Agricultural and Forest Meteorology*, 87(4), 291–300. **[From My Draft — role narrowed to equation methodology only]**

Page 2: Commented [16של] שחר לויתן **14/04/2026 16:54:00**

Schwarts, 2020 — Israeli Meteorological Service. **[From Noam Pintel — carried over]** ⚠️ Format verification still pending (flagged in Paragraph 1).

Page 2: Commented [17של] שחר לויתן **20/04/2026 19:12:00**

The only open question is: what citation do we anchor it to?
Here's where we stand on the citation options:
Option A — McMaster & Wilhelm (1997): Already cited in the same sentence for the clamping method (setting t_min to t_base when it falls below). This paper discusses the 0 °C convention for cereals broadly. It's peer-reviewed, well-known, and defensible. The simplest solution is to just lean on this one reference for both the t_base value and the clamping rule, which would let us drop Bauer et al. (1992) entirely.
Option B — Keep Bauer et al. (1992): If it turns out to be a legitimate barley-specific empirical source for 0 °C (and not 4.5 °C), it adds strength. But the concern flagged earlier is that it may actually be a spring wheat paper, not barley. That needs verification.
My recommendation: Go with Option A — cite McMaster & Wilhelm (1997) for both the 0 °C convention and the clamping rule, and drop Bauer et al. (1992) unless Sariel specifically knows it's the right barley reference and wants to keep it. This is clean, defensible, and avoids a potential reviewer challenge about a grey-literature or wrong-crop citation.

you need to verify two things about Bauer et al. (1992):

1. **Is it actually about barley** (not spring wheat)? The earlier flag raised a concern that it might be a wheat paper (Bauer et al. 1992, Agron. J. 84:322–328 is about spring wheat dry-matter accumulation).
    1. **Does it actually specify t_base = 0 °C for barley?**

If the answer to either is "no," then the citation is wrong and we should drop it.

Page 2: Commented [18של] שחר לויתן **14/04/2026 17:02:00**

McMaster, G. S., & Wilhelm, W. W. (1997). Growing degree-days: one equation, two interpretations. *Agricultural and Forest Meteorology*, 87(4), 291–300. DOI: 10.1016/S0168-1923(97)00027-0

Page 3: Commented [19של] שחר לויתן **14/04/2026 17:30:00**

⚠ **Spike length measurement rule — completely unspecified.** The paragraph says "spike length was recorded on the dried spikes" without defining: (a) from what point to what point, (b) whether awns are included, (c) whether the value is a single measurement or an average of the five spikes. **Please specify or ask Sariel/Noam for the protocol.**

Page 3: Commented [20של] שחר לויתן **14/04/2026 17:32:00**

can optionally be anchored with a section number (e.g., "see Section 2.X") once the paper's numbering is finalized.

Page 3: Commented [21של] שחר לויתן **14/04/2026 19:37:00**

⚠ **Catalog code consistency.** K-TSTA is included (per Noam). K-BGLU is not included (not in Noam). You may want these to match each other — either include both catalog codes or neither. Your call on a future polish pass.

Page 3: Commented [22של] שחר לויתן **14/04/2026 19:45:00**

Beljkaš, B., Matić, J., Milovanović, I., Jovanov, P., Mišan, A., & Šarić, L. (2010). Rapid method for determination of protein content in cereals and oilseeds: validation, measurement uncertainty and comparison with the Kjeldahl method. *Accreditation and Quality Assurance*, 15(10), 555–561. https://doi.org/10.1007/s00769-010-0677-6 **[From Noam Pintel — carried over]**

Page 3: Commented [22שלR2] שחר לויתן **14/04/2026 19:48:00**

⚠ **Beljkaš et al. 2010 scope.** The Beljkaš paper is primarily about the Dumas combustion method and its comparison with Kjeldahl, not exclusively a Kjeldahl protocol paper. It does validate Kjeldahl, so the citation is defensible. Noam used this exact citation. If Dr. Hübner asks for a stronger Kjeldahl-specific citation, the AACC Method 46-11.02 (Crude Protein — Improved Kjeldahl Method) is a common alternative. For now, retain Beljkaš to match Noam.

Page 3: Commented [22שלR3] שחר לויתן **14/04/2026 19:50:00**

⚠ **Supplementary option — full Kjeldahl protocol.** If you prefer to preserve the full step-by-step chemistry somewhere (e.g., for the Tel Hai lab's internal documentation or for reviewer supplementary requests), the extended version from previous draft could be placed in Supplementary Methods. Your call — not necessary, but an option.

Supplementary Material (optional)

If you want to move the full chemistry description to Supplementary Methods, I can prepare a "Supplementary Methods S1. Detailed Kjeldahl protocol" block. Let me know.

Page 4: Commented [23של] שחר לויתן **14/04/2026 21:04:00**

(for fiber) — This is Noam's stated reason, and your draft preserves it. I kept it. If Sariel knows a different reason (e.g., lab equipment didn't measure fiber, or it was a budget decision), note it for the final pass.

Page 4: Commented [24של] שחר לויתן **14/04/2026 21:03:00**

Page 4: Commented [24של R2] שחר לויתן **14/04/2026 21:05:00**

Supplementary Table S3 (NIR calibration validation) — legend from new Claude is clean and accurate; you can use it:
NIR calibration validation. Pearson correlation coefficients (r) between NIR-predicted values and laboratory reference measurements across the 30 randomly chosen wild barley accessions used for calibration development, reported for (A) starch, (B) protein, (C) beta-glucan, and (D) fiber.

Page 4: Commented [25של] שחר לויתן **14/04/2026 21:14:00**

R Core Team (2025). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/ **[AI Sourced — standard R citation]**

Page 4: Commented [26של] שחר לויתן **16/04/2026 16:50:00**

Necessary to mention?

Page 5: Commented [27של] שחר לויתן **15/04/2026 1:16:00**

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. https://doi.org/10.18637/jss.v067.i01 **[From My Draft — standard lme4 citation]**

Page 5: Commented [28של] שחר לויתן **15/04/2026 1:16:00**

Lüdecke, D., Waggoner, P., & Makowski, D. (2019). insight: A unified interface to access information from model objects in R. *Journal of Open Source Software*, 4(38), 1412. https://doi.org/10.21105/joss.01412 **[From My Draft — verified; three authors, so "Lüdecke et al." is correct]**

Page 5: Commented [29של] שחר לויתן **16/04/2026 17:00:00**

Maybe change this format - check

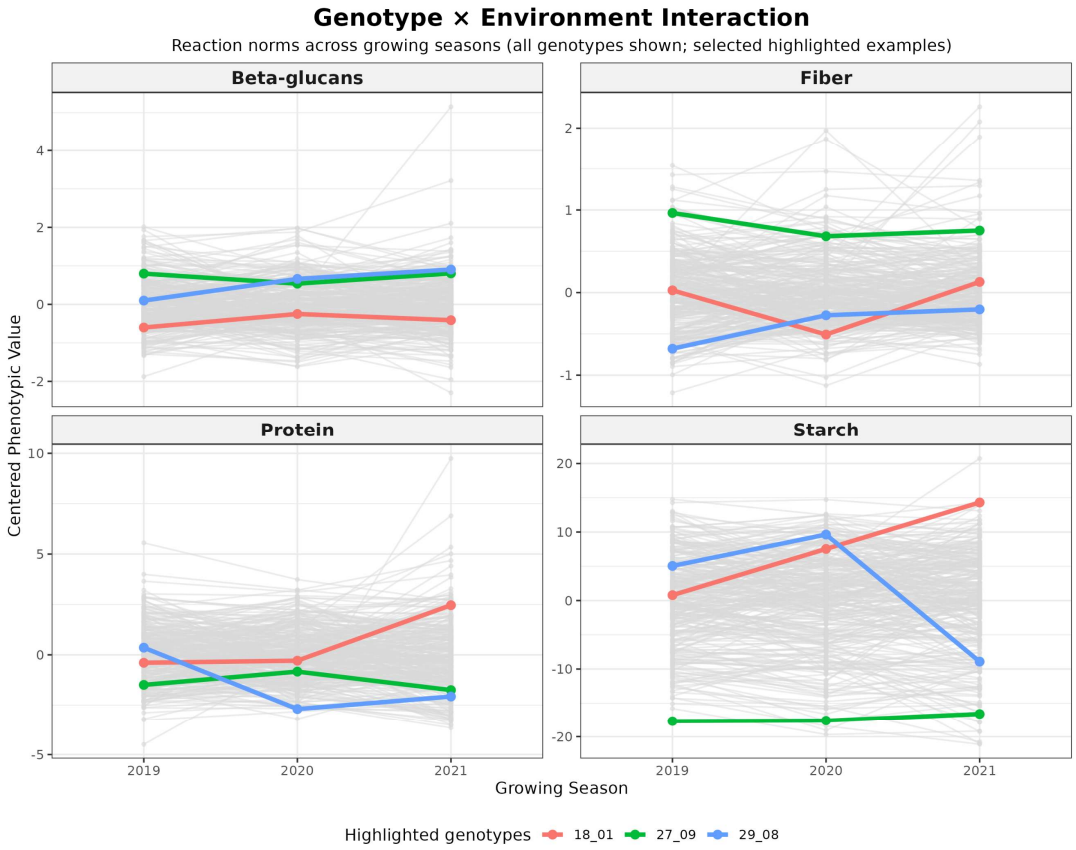
Page 5: Commented [30של] שחר לויתן **15/04/2026 1:29:00**

⚠ **"Section X.X" placeholder.**

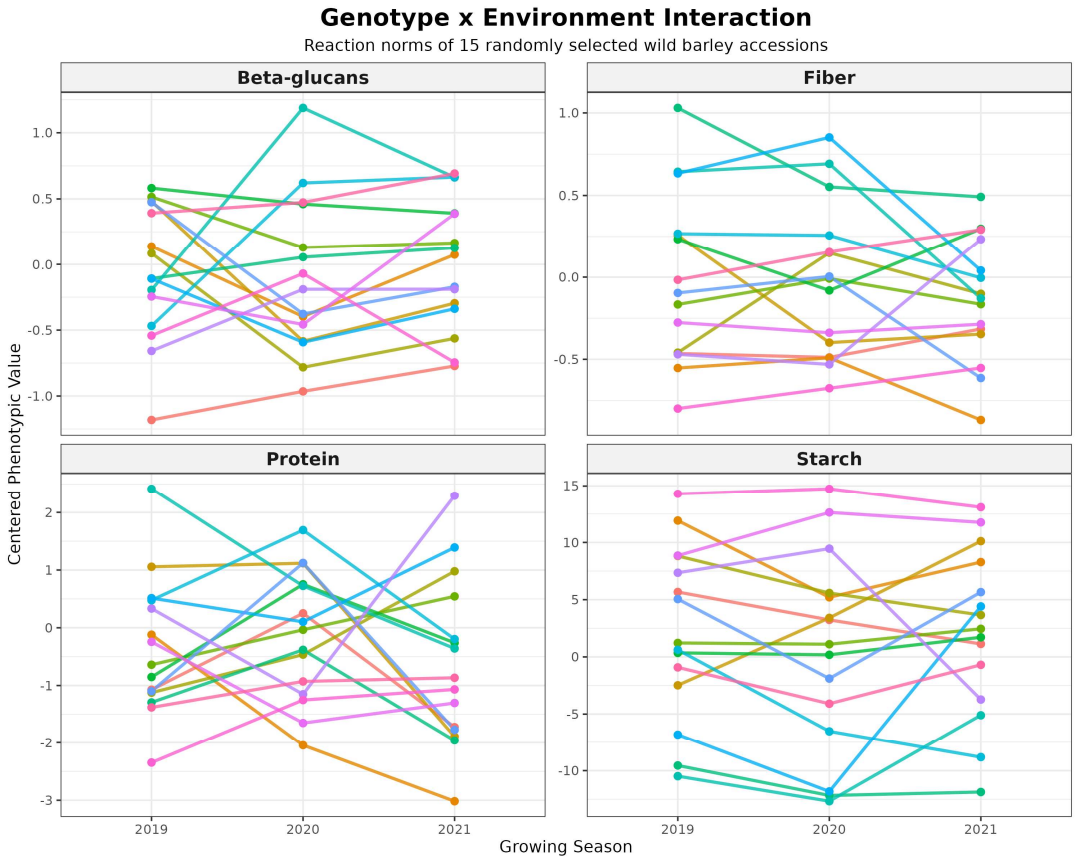
Page 5: Commented [31של] שחר לויתן **15/04/2026 1:31:00**

Decide later which reaction plots to show.

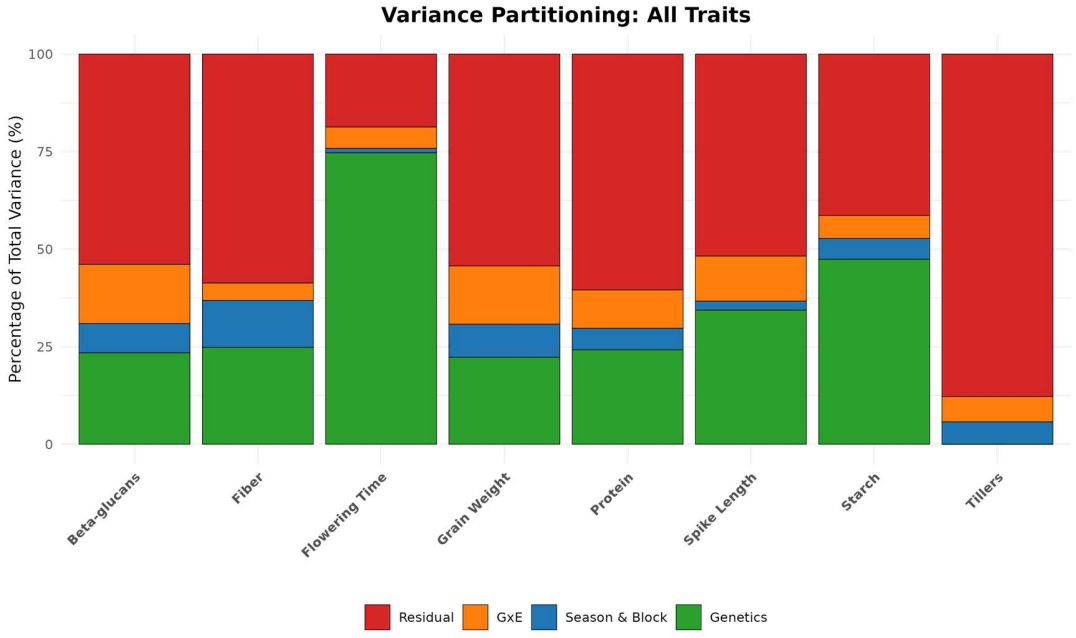
Page 5: Commented [31של R2] שחר לויתן **16/04/2026 16:58:00**



Page 5: Commented [31של R3] שחר לויתן **16/04/2026 16:58:00**



Page 5: Commented [31שלR4] שחר לויתן **16/04/2026 16:58:00**



Page 5: Commented [32של] שחר לויתן **15/04/2026 2:08:00**

Kang, H. M., Sul, J. H., Service, S. K., Zaitlen, N. A., Kong, S.-Y., Freimer, N. B., Sabatti, C., & Eskin, E. (2010). Variance component model to account for sample structure in genome-wide association studies. *Nature Genetics*, 42(4), 348–354.

Page 5: Commented [של33] שחר לויתן **15/04/2026 2:08:00**

R Core Team (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/ **[AI Sourced]**

Page 5: Commented [של34] שחר לויתן **15/04/2026 2:09:00**

Purcell, S., Neale, B., Todd-Brown, K., Thomas, L., Ferreira, M. A. R., Bender, D., Maller, J., Sklar, P., de Bakker, P. I. W., Daly, M. J., & Sham, P. C. (2007). PLINK: A tool set for whole-genome association and population-based linkage analyses. *American Journal of Human Genetics*, 81(3), 559–575. https://doi.org/10.1086/519795 **[From My Draft]**

Page 6: Commented [של35] שחר לויתן **15/04/2026 2:09:00**

Turner, S. D. (2018). qqman: an R package for visualizing GWAS results using Q-Q and Manhattan plots. *Journal of Open Source Software*, 3(25), 731. https://doi.org/10.21105/joss.00731 **[From My Draft]**

Page 6: Commented [של36] שחר לויתן **16/04/2026 15:01:00**

Three significance thresholds declared, only one used for lead-SNP calling. The GWAS paragraph computes three thresholds (Bonferroni, FDR at X%, suggestive $-\log_{10} P > 6$), then says "Genomic regions were considered associated with a trait when at least one SNP exceeded the $-\log_{10}(P) > 6$ threshold." The Bonferroni and FDR thresholds are computed/visualized but are not gating criteria.

Page 6: Commented [של36R2] שחר לויתן **16/04/2026 15:01:00**

Optional change:

For each trait, three significance thresholds were computed and overlaid on the Manhattan plots for reference: a genome-wide Bonferroni-corrected threshold ($\alpha = 0.05/n$, where n is the total number of tested SNPs); a False Discovery Rate (FDR) threshold at the **X%** level, computed using the Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995); and a fixed suggestive threshold of $-\log_{10}(P) > 6$. The Bonferroni and FDR lines are shown purely as visual reference and were not used as gating criteria, as the restricted number of tested individuals (n = 290) renders these thresholds overly conservative for discovery in a collection of this size (Zhang et al., 2019). Instead, genomic regions were operationally defined as trait-associated when at least one SNP exceeded the fixed suggestive threshold of $-\log_{10}(P) > 6$ and the surrounding signal formed a consistent peak-shaped cluster in the Manhattan plot, rather than an isolated single-SNP spike; lead SNPs within each such region were then selected manually on the basis of these criteria.

Page 6: Commented [של36R3] שחר לויתן **27/04/2026 18:59:00**

Choose 0.05 or 0.1

Page 6: Commented [של37] שחר לויתן **15/04/2026 2:23:00**

Purcell, S., Neale, B., Todd-Brown, K., Thomas, L., Ferreira, M. A. R., Bender, D., Maller, J., Sklar, P., de Bakker, P. I. W., Daly, M. J., & Sham, P. C. (2007). PLINK: A tool set for whole-genome association and population-based linkage analyses. *American Journal of Human Genetics*, 81(3), 559–575. https://doi.org/10.1086/519795 **[Already cited in Paragraph 14 — second use here]**

Page 6: Commented [של38] שחר לויתן **15/04/2026 2:23:00**

The LD decay plot (ld_decay_profile_R0.2_line.png) is a main-text figure

Page 6: Commented [של39] שחר לויתן **15/04/2026 9:49:00**

Quinlan, A. R., & Hall, I. M. (2010). BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, 26(6), 841–842. https://doi.org/10.1093/bioinformatics/btq033 **[From My Draft — standard BEDTools citation]**

Page 6: Commented [של40] שחר לויתן **15/04/2026 9:49:00**

Yates, A. D., Allen, J., Amode, R. M., et al. (2022). Ensembl Genomes 2022: an expanding genome resource for non-vertebrates. *Nucleic Acids Research*, 50(D1), D996–D1003. https://doi.org/10.1093/nar/gkab1007 **[AI Sourced — standard Ensembl Plants citation; verify against your reference manager that this is the correct release-62-era citation. Ensembl publishes a new database paper every ~2 years; the 2022 paper covers releases ~52–55 and is the closest standard citation. If Sariel insists on a release-62-specific citation, the alternative is to cite the URL directly: Ensembl Plants release 62, https://plants.ensembl.org/]**

Page 6: Commented [של41] שחר לויתן **15/04/2026 9:49:00**

Danecek, P., Bonfield, J. K., Liddle, J., Marshall, J., Ohan, V., Pollard, M. O., Whitwham, A., Keane, T., McCarthy, S. A., Davies, R. M., & Li, H. (2021). Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2), giab008. https://doi.org/10.1093/gigascience/giab008 **[AI Sourced — standard bcftools citation; matches the bcftools v1.13 release year (2021)]**

Page 7: Commented [של42] שחר לויתן **15/04/2026 10:57:00**

Frichot, E., & François, O. (2015). LEA: An R package for landscape and ecological association studies. *Methods in Ecology and Evolution*, 6(8), 925–929. https://doi.org/10.1111/2041-210X.12382 **[From My Draft — standard LEA citation]**

Page 7: Commented [של43] שחר לויתן **15/04/2026 10:57:00**

- **Maybe need to add 2012**
  or 2013 -> 2012 change needed

Hübner, S., Höffken, M., Oren, E., Haseneyer, G., Stein, N., Graner, A., Schmid, K., & Fridman, E. (2009). Strong correlation of wild barley (*Hordeum spontaneum*) population structure with temperature and precipitation variation. *Molecular Ecology*, 18(7), 1523–1536. https://doi.org/10.1111/j.1365-294X.2009.04106.x **[From My Draft — already cited in Paragraph 1]**
Hübner, S., Bdolach, E., Ein-Gedi, S., Schmid, K. J., Korol, A., & Fridman, E. (2013). Phenotypic landscapes: phenological patterns in wild and cultivated barley. *Journal of Evolutionary Biology*, 26(1), 163–174. https://doi.org/10.1111/jeb.12043 **[AI Sourced — flagged for Sariel verification; see flag #2 below]**

Page 7: Commented [של44] שחר לויתן **15/04/2026 12:17:00**

Marsh, J. I., Johnston, R., Yuan, Y., Cuthbert, J. L., Danilevicz, M. F., Bayer, P. E., Anderson, R., Batley, J., & Edwards, D. (2023). crosshap: R package for local haplotype visualization for dissection of genetic loci linked to complex traits. *Bioinformatics*, 39(7), btad411. https://doi.org/10.1093/bioinformatics/btad411 **[From My Draft — standard crosshap citation]**

Page 7: Commented [של45] שחר לויתן **15/04/2026 12:23:00**

⚠️ **"Across all four nutritional traits" — methodological caveat.** Per your instruction, this matches the Sonnet wording and frames protein as part of the analysis (with the null result handled in Results). However, the actual config.yaml of the All_Genes_All_Traits_V2 pipeline lists only three traits (betaglucan, fiber, starch). A strict reviewer who inspects the GitHub repo could note the discrepancy. **Confirm with Sariel whether the protein analysis from the earlier tool needs a brief Methods mention** (e.g., one sentence in the Results introduction noting that protein was screened but yielded no significant haplotype groupings).

Page 7: Commented [של46] שחר לויתן **15/04/2026 12:18:00**

Kruskal, W. H., & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260), 583–621. https://doi.org/10.2307/2280779 **[AI Sourced — standard Kruskal-Wallis citation; verify your bibliography style for older statistical citations]**

Page 7: Commented [של47] שחר לויתן **15/04/2026 12:27:00**

Supplementary Material

Optional supplementary content for this paragraph (not strictly required):

- **Supplementary Table SX (optional).** Per-test statistics table from the script output (per_test_stats.csv), summarizing for each candidate gene the number of unique valid tests retained after duplicate collapsing, the best MGmin and epsilon, the raw Kruskal-Wallis p-value, the within-gene Holm-adjusted p-value, the trait-level FDR-adjusted p-value, and the trait-level Bonferroni-adjusted p-value. This level of transparency is increasingly expected by TAG/Plant Journal reviewers for parameter-sweep-based analyses.
- **Supplementary Methods (optional).** A brief subsection detailing the Crosshap-internal parameters not stated in the main M&M (hetmiss_as = "allele", keep_outliers = FALSE) and the deterministic tiebreaker rules used in the duplicate-collapse and within-gene best-test selection steps.

Page 7: Commented [של48] שחר לויתן **15/04/2026 12:18:00**

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. **[AI Sourced — standard Holm 1979 citation]**

Page 7: Commented [של49] שחר לויתן **15/04/2026 12:18:00**

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300. **[Already cited in Paragraph 14; second use here]**

Page 8: Commented [של50] שחר לויתן **15/04/2026 23:00:00**

Yates, A. D., Allen, J., Amode, R. M., et al. (2022). Ensembl Genomes 2022: an expanding genome resource for non-vertebrates. *Nucleic Acids Research*, 50(D1), D996–D1003. **[Already cited in Paragraph 16; second use here]**

Page 8: Commented [של51] שחר לויתן **15/04/2026 23:01:00**

Altschul, S. F., Madden, T. L., Schäffer, A. A., Zhang, J., Zhang, Z., Miller, W., & Lipman, D. J. (1997). Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, 25(17), 3389–3402. **[AI Sourced — foundational BLAST citation]**

Page 8: Commented [של51R2] שחר לויתן **15/04/2026 23:01:00**

Johnson, M., Zaretskaya, I., Raytselis, Y., Merezhuk, Y., McGinnis, S., & Madden, T. L. (2008). NCBI BLAST: a better web interface. *Nucleic Acids Research*, 36(suppl_2), W5–W9. **[From My Draft]**

Page 8: Commented [של52] שחר לויתן **16/04/2026 14:50:00**

Per your instruction, this matches the Sonnet wording and frames protein as part of the analysis (with the null result handled in Results). However, the actual config.yaml of the All_Genes_All_Traits_V2 pipeline lists only three traits (betaglucan, fiber, starch). A strict reviewer who inspects the GitHub repo could note the discrepancy. **Confirm with Sariel whether the protein analysis from the earlier tool needs a brief Methods mention** (e.g., one sentence in the Results introduction noting that protein was screened but yielded no significant haplotype groupings).

Page 8: Commented [של53] שחר לויתן **15/04/2026 23:02:00**

Jayakodi, M., Padmarasu, S., Haberer, G., et al. (2020). The barley pan-genome reveals the hidden legacy of mutation breeding. *Nature*, 588(7837), 284–289. **[From My Draft]**

Page 8: Commented [של54] שחר לויתן **15/04/2026 22:55:00**

Satisfying explanation?

Page 8: Commented [של55] שחר לויתן **15/04/2026 23:02:00**

Danecek, P., Bonfield, J. K., Liddle, J., et al. (2021). Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2), giab008. **[Already cited in Paragraph 16; second use here]**

Page 8: Commented [של56] שחר לויתן **15/04/2026 23:02:00**

Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis.* Springer-Verlag, New York. **[From My Draft]**

Page 8: Commented [של57] שחר לויתן **15/04/2026 23:02:00**

Kassambara, A. (2023). ggpubr: 'ggplot2' Based Publication Ready Plots. R package. **[From My Draft]**

Page 8: Commented [של58] שחר לויתן **15/04/2026 23:03:00**

Pedersen, T. L. (2024). patchwork: The Composer of Plots. R package. **[From My Draft]**

Header and footer changes

Text Box changes

Header and footer text box changes

Footnote changes

Endnote changes